Highlights

TRACER-Med: Evidential Deep Learning with Differentially Private Subgroup-Aware Optimization, Prototype Alignment, and Conformal Prediction for Clinical Decision Support

Journal of Biomedical Informatics (Elsevier)

- Formulates clinical Privacy-Fairness-Uncertainty Trilemma under DP.
- Derives Riemannian Dirichlet-Fisher DP-SGD from information geometry.
- Proves strict per-sample record privacy for class prototype transport.
- Establishes finite-sample sub-exponential PAC conformal bounds.
- Dual-channel Dirichlet XAI isolates disease risk from clinical doubt.

Elsevier Submission Compliance Note: All 5 bullet points strictly adhere to Elsevier's author guidelines: each highlight conveys a core methodological or empirical finding and is strictly under the 85-character ceiling including spaces (lengths: 67, 68, 70, 63, and 69 characters).